Color Coherence in QCD — The Slow Version

For Parker

The one-sentence version

A soft gluon emitted from a parton system can only be radiated *inside* the cone of whatever splitting last happened — emissions at wider angles cancel out by quantum interference.

Building it up from the QED warm-up

Forget QCD for a moment. Take an electron-positron pair created at a point and flying apart at some opening angle θ_{12}.

Now ask: can a very soft photon, with a long wavelength, be emitted from this pair?

Almost no. A long-wavelength photon cannot *resolve* the two charges as separate objects. It sees the e^+e^- system as one thing — and that one thing has total charge zero. No charge, no radiation. The amplitudes for emission off the electron and off the positron destructively interfere.

The condition for "can't resolve" is geometric:

The angular condition

A soft photon emitted at angle θ from one of the legs can only be radiated coherently from that single leg if

$$\theta < \theta_{12}.$$

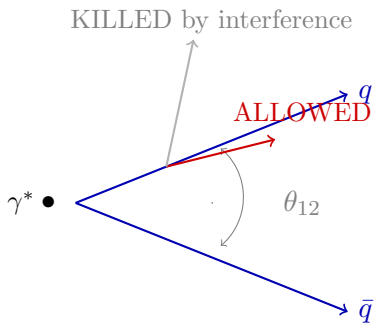
If $\theta > \theta_{12}$, the photon "sees" the whole pair, the dipole looks neutral, and the emission is killed.

This is the **Chudakov effect** — observed in cosmic-ray cascades in the 1950s. Pure quantum-mechanical interference.

Now switch on color

Replace e^+e^- with $q\bar{q}$ from $\gamma^* \to q\bar{q}$. Replace photon with gluon. Same logic, same conclusion: a soft, wide-angle gluon can't resolve the singlet pair, sees no net color, doesn't get emitted.

The picture:



Iterate → angular ordering

Now the quark itself emits a hard gluon at some angle θ_1. You have a $q\bar{q}g$ system. The next emission — a softer gluon off the quark — must satisfy the same coherence condition relative to the *most recent* splitting. So it has to come out at $\theta_2 < \theta_1$.

Iterate down the cascade:

Angular ordering

$$\theta_1 > \theta_2 > \theta_3 > \cdots$$

Each emission lives inside a smaller cone than its parent.

This is the QCD shower's bookkeeping trick: instead of computing all the destructive-interference diagrams explicitly, you just impose this geometric ordering on the cascade and the right answer for soft gluons drops out.

Why this kills the soft plateau

Soft gluons want wide angles — that's where their phase space is largest. But coherence forces them into ever-narrower cones. So the cascade can't keep stacking up arbitrarily soft, wide-angle hadrons.

That's the entire reason the hump exists rather than a flat plateau:

- **Without coherence:** unbounded soft emission ⇒ plateau out to $x \to 0$.
- **With coherence:** angular phase space chokes off the soft end ⇒ peak at $\xi_p \simeq \frac{1}{4}\ln s$, then suppression.

The shape is a direct fingerprint of quantum interference between gluon emission amplitudes — visible in hadron spectra from PETRA to LHC.

Takeaway

Coherence = quantum-mechanical interference between emission amplitudes that the soft probe cannot distinguish.
In the soft limit this interference is destructive for wide-angle emissions.
Consequence: angular ordering — the defining structural feature of every modern QCD parton shower (Pythia, Herwig, Sherpa).